

29. FORWARD CHAINING

AIM

To develop a Prolog program that implements forward chaining by deriving new facts from existing facts and predefined rules.

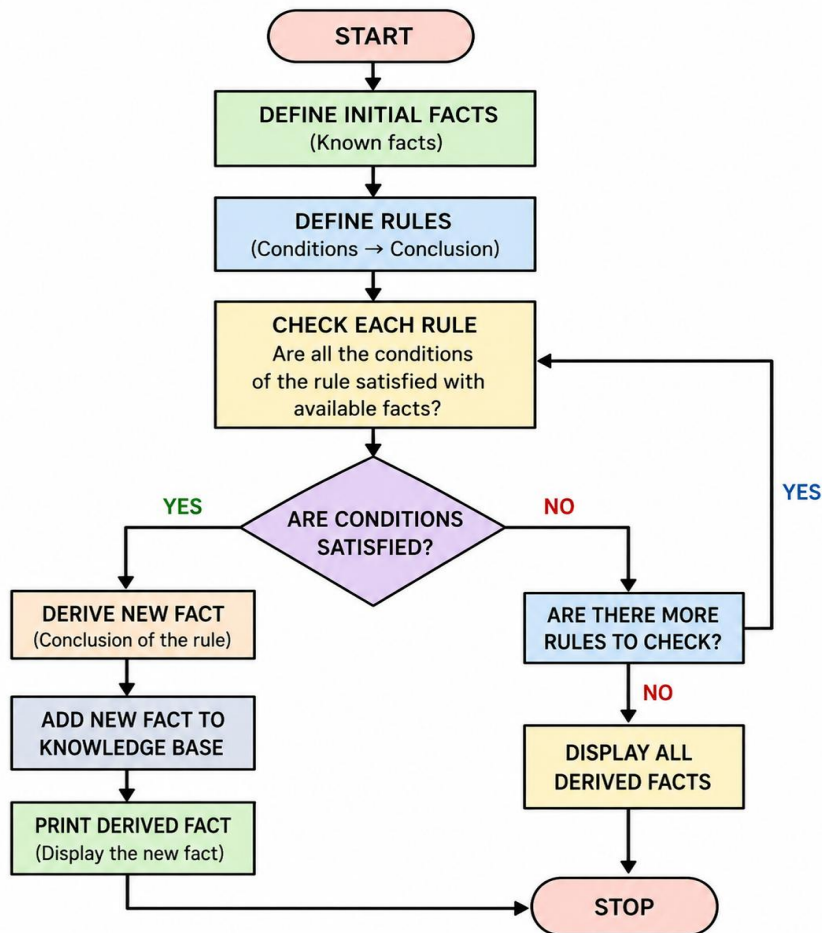
OBJECTIVE

- To understand the concept of forward chaining in Artificial Intelligence.
- To represent knowledge using facts and rules in Prolog.
- To derive new facts automatically from existing facts.
- To execute the required queries and verify the derived conclusions.

ALGORITHM

1. Start the program.
2. Define the initial facts.
3. Define rules using the available facts.
4. Check whether the conditions of a rule are satisfied.
5. If the conditions are satisfied, derive a new fact.
6. Add the newly derived fact to the knowledge base.
7. Continue checking the rules until no new facts can be derived.
8. Display the derived facts.
9. Stop the program.

FLOWCHART



CODE

% FORWARD CHAINING PROGRAM

:- dynamic fact/1.

% INITIAL FACTS

fact(fever).

fact(cough).

fact(body_pain).

% RULES

rule(fever, cough, flu).

rule(cough, body_pain, infection).

rule(flu, infection, serious_flu).

% FORWARD CHAINING

forward_chain :-

```
rule(A, B, C),  
fact(A),  
fact(B),  
\+ fact(C),  
assertz(fact(C)),  
write('Derived fact: '),  
write(C),  
nl,  
fail.
```

forward_chain.

% START PROGRAM

start :-

```
forward_chain,  
write('Forward chaining completed.'). nl.
```

OUTPUT

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)  
File Edit Settings Run Debug Help  
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.  
  
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).  
  
?-  
% c:/Users/New/Documents/Prolog/forward chaining.pl compiled 0.00 sec, 9 clauses  
?- fact(fever).  
true.  
  
?- fact(cough).  
true.  
  
?- start.  
Derived fact: flu  
Derived fact: infection  
Derived fact: serious_flu  
Forward chaining completed.  
true.  
?-
```

RESULT

Thus, the forward chaining program was successfully implemented in Prolog. The program starts with known facts, applies the predefined rules, derives new facts, and continues the process until all possible conclusions are obtained.